

KARUNASAGAR GUDAS

Senior Special Process Engineer | Aerospace Manufacturing & NADCAP Compliance

📍 Dharwad, Karnataka – 580001 | 📞 +91 87924 79309 | ✉️ agkarunasagar@gmail.com

PROFESSIONAL SUMMARY

Results-oriented Special Process Engineer with **7+ years** of hands-on experience in the **aerospace industry**, specializing in **NDT, Heat Treatment and Surface Treatments**. Recognized for successfully leading **NADCAP certifications**, process industrialization, and compliance initiatives for major OEMs like **Airbus, Collins, Boeing, Eaton, and Triumph**. Expert in **AMS2750 pyrometry**, process audits, technical documentation, and cross-functional collaboration. Proven ability to enhance operational efficiency, ensure regulatory compliance, and mentor high-performing technical teams.

PROFESSIONAL EXPERIENCE

Senior Special Process Engineer – Process Engineering

Unitritech Pvt. Ltd., Dharwad, Karnataka

📅 April 2021 – Present

- Manage Heat Treatment and NDT process lines ensuring compliance and output.
- Develop process documentation: Manufacturing Instructions, Control Plans, Procedures, and Training Modules.
- Execute pyrometry activities in compliance with **AMS2750**.
- Heat Treatment to T4 T6 condition for non ferrous Casting as per the requirements of Industrial standards.
- Calibration of Heat Treatment furnace as per AMS 2750 in compliance with NADCAP requirements.
- Review and calibrate gauges/equipment used in NDT, Heat Treatment, and Welding.
- Conduct annual performance reviews for operators and Level I/II NDT personnel per NADCAP and QMS.
- Handle contract reviews based on technical/customer data (Airbus, Collins USA/Germany, Triumph, Eaton).
- Coordinate with cross-functional teams for new product industrialization and process improvements.
- Interface with suppliers for audits, calibration, and process approvals.
- Resolved process non-conformities through structured **root cause analysis, CAPA, PFMEA** and 5s

Key Achievements:

- **NADCAP Certification (Zero Major NCs):**
 - NDT – 2023-24 (Audit No. 224769)
 - NDT – 2022-23 (Audit No. 207647)
 - Heat Treatment – 2021-22 (Audit No. 201840)

Trained Level I technicians to qualify as Level II in NDT.

Special Process Engineer – Process Engineering

Lakshmi Machine Works Ltd. – Advanced Technology Centre, Coimbatore

■ Jan 2021 – Apr 2021

- Managed **Surface and Heat Treatment** processes for aerospace clients including **Honeywell, Boeing, and Dassault Aviation**.
- Oversaw **chemical processing lines**: Chromic Acid Anodizing, BSAA, Hard Anodizing, and Chemical Milling.
- Ensured **pyrometry compliance**, audit readiness, and production target alignment.
- Supported customer approvals, job clearances, and resource planning.

Special Process Engineer – QA (NAS410/EN4179 FPI Level 2)

Unitritech India Pvt. Ltd. – A Division of Chemical & Ferro Alloys Pvt. Ltd.

■ Nov 2018 – Dec 2020

- Administered **NDT and Heat Treatment** for investment castings in aerospace applications.
- Performed inspections in compliance with **ABP, AITM, ASTM, MIL-STD** for clients such as **Boeing, Airbus, and UTAS**.
- Maintained **bath quality**, executed pyrometry tasks, and handled equipment calibration.
- Drafted and controlled **technical documentation**, including process sheets and quality plans.
- Resolved process non-conformities through structured **root cause analysis and CAPA**.

Key Achievements:

- NADCAP Certification – FPI & Chemical Etching:
 - 2020-21 (Audit No. 196295)
 - 2018-19 (Audit No. 185242)
- Successfully aligned documentation for NADCAP compliance (2018–2019).

CORE COMPETENCIES

- **Special Processes**: NDT (FPI, MPI), Heat Treatment, Welding, Surface Treatments
- **Certifications & Compliance**: NADCAP Audits, NAS410 / EN4179, AMS2750
- **Process Engineering**: Control Plans, SOPs, Troubleshooting, PFMEA
- **Quality Management**: QMS, CAPA, Internal Audits, Root Cause Analysis
- **Lean Manufacturing**: 5S, Kaizen, Continuous Improvement
- **Technical Leadership**: Team Training, Operator Qualification, Cross-functional Coordination
- **Customer Interface**: Airbus, Collins (USA/Germany), Boeing, Triumph, Eaton

CERTIFICATIONS

- **NAS410 / EN4179 Level II** – Fluorescent Penetrant (FPI) & Magnetic Particle (MPI)
- **NADCAP Audit Proficiency** – NDT, Heat Treatment, Surface Treatment
- Proficient in **AMS2750** pyrometry standards

EDUCATION

Bachelor of Engineering – Mechanical Engineering

Jain College of Engineering, Belgaum, Karnataka

🎓 Graduated: 2015

LANGUAGES

- **English** – Full Professional Proficiency
- **Kannada** – Native Proficiency
- **Hindi** – Professional Working Proficiency
- **Marathi** – Conversational Proficiency

DECLARATION

I hereby declare that the above-mentioned information is correct to the best of my knowledge. I bear responsibility for the correctness of the above-mentioned particulars.